

BERGARUD, Sweden

WMO: 025210

Lat: 58.6003N

Lon: 12.4130E

Elev: 194

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.3	(c) 8.9	(d) -0.6	(e) 5.3	(f) 2.6	(g) 5.3	(h) 7.3	(i) 10.2	(j) 29.2	(k) 45.6	(l) 25.8	(m) 42.0	(n) 1.6	(o) 320	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.6	(c) 79.1	(d) 65.1	(e) 75.9	(f) 63.4	(g) 72.8	(h) 61.7	(i) 67.2	(j) 74.6	(k) 65.3	(l) 72.3	(m) 63.7	(n) 70.3	(o) 7.8	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.7	(b) 92.1	(c) 69.7	(d) 62.9	(e) 86.3	(f) 67.8	(g) 61.1	(h) 81.0	(i) 66.1	(j) 31.7	(k) 75.2	(l) 30.3	(m) 72.3	(n) 29.1	(o) 70.5	(p) 73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.4	17.3	-6.9	85.3	8.6	2.9	-13.1	87.4	-18.1	89.1	-22.9	90.7	-29.2	92.8
			-7.3	70.4	8.0	1.7	-13.0	71.5	-17.7	72.5	-22.2	73.4	-28.0	74.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	29.2	29.6	34.6	43.3	51.3	57.9	62.5	61.4	54.0	45.6	38.9	32.3		
	DBStd	13.62	9.76	8.58	7.64	5.52	5.34	4.62	4.26	4.23	5.03	6.78	7.19	10.08		
	HDD50	3028	646	571	480	211	45	2	0	0	21	168	336	548		
	HDD65	7295	1111	991	944	650	424	218	102	126	332	600	784	1013		
	CDD50	1253	0	0	1	11	86	238	389	353	140	32	3	0		
	CDD65	45	0	0	0	0	0	4	26	14	1	0	0	0		
	CDH74	467	0	0	0	0	10	106	232	115	4	0	0	0		
	CDH80	53	0	0	0	0	1	13	31	8	0	0	0	0		
(14) Wind	WSAvg	8.1	9.6	8.6	8.3	8.2	8.5	7.3	6.8	7.0	8.2	7.6	8.7	8.4		
(15) Precipitation	PrecAvg	32.6	2.7	2.2	1.7	2.0	2.0	3.1	3.1	3.3	3.2	3.4	3.1	2.9		
	PrecMax	44.7	5.1	5.0	3.9	4.2	3.3	5.0	7.6	6.7	8.4	7.8	5.7	6.1		
	PrecMin	24.4	0.4	0.4	0.3	0.3	0.4	1.1	0.3	1.0	0.8	0.8	1.0	0.5		
	PrecStd	5.0	1.3	1.2	0.9	1.3	0.9	1.3	1.7	1.4	2.0	1.7	1.4	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	47.5	57.6	68.5	74.6	81.6	84.1	81.0	73.3	62.4	54.8	51.1		
		MCWB	44.6	43.3	47.8	54.6	57.9	64.8	67.5	64.7	60.0	57.3	52.2	49.4		
	2%	DB	44.8	44.6	51.7	62.6	70.0	76.9	79.8	77.4	68.7	58.9	52.7	48.5		
		MCWB	42.9	41.2	43.5	51.3	57.2	62.2	66.1	64.3	57.9	55.1	50.6	46.7		
	5%	DB	42.7	42.3	47.3	58.0	65.8	72.8	76.7	74.0	65.6	57.0	51.1	46.7		
		MCWB	40.8	40.1	41.5	48.2	54.9	60.5	64.5	62.9	57.2	53.9	49.0	44.7		
	10%	DB	41.0	40.6	44.0	54.1	61.6	69.3	73.3	70.7	62.9	55.1	49.1	44.6		
		MCWB	39.3	38.6	40.3	46.2	52.9	58.9	63.1	61.3	56.1	52.6	47.2	42.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	45.6	48.1	55.5	61.9	67.2	70.2	68.1	62.2	58.5	53.1	49.8		
		MCDB	46.6	46.9	57.1	64.6	70.1	77.4	78.6	74.5	67.5	60.7	54.3	50.9		
	2%	WB	43.2	42.1	44.8	52.4	58.6	64.4	68.2	66.0	60.2	56.6	51.0	47.3		
		MCDB	44.4	43.2	49.6	60.8	64.8	73.8	76.1	71.9	65.0	58.2	52.4	48.4		
	5%	WB	41.3	40.4	42.7	49.6	56.6	62.0	66.3	64.6	58.8	54.6	49.4	45.1		
		MCDB	42.5	42.0	46.1	56.2	63.5	70.2	73.4	70.9	63.2	56.3	51.0	46.6		
	10%	WB	39.4	38.6	40.9	47.2	54.2	59.9	64.5	63.2	57.4	53.0	47.3	42.7		
		MCDB	40.9	40.5	43.6	52.8	60.4	67.6	71.3	69.1	61.3	55.0	49.0	44.2		
(35) Mean Daily Temperature Range	MDBR		8.9	9.8	12.8	16.3	16.2	17.3	16.6	16.5	15.4	11.8	8.4	8.5		
	5% DB	MCDBR	9.3	9.3	17.3	24.2	22.9	25.1	23.7	22.2	19.9	12.3	9.2	7.4		
		MCWBR	9.0	7.2	11.2	13.8	11.5	12.1	11.0	10.4	10.9	8.8	8.4	7.5		
	5% WB	MCDBR	9.2	8.4	14.8	21.6	18.7	22.0	19.9	17.7	16.1	10.5	8.6	7.1		
		MCWBR	8.9	7.1	10.3	12.9	10.5	11.8	10.7	9.7	10.2	8.6	8.2	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.315	0.336	0.374	0.370	0.359	0.380	0.376	0.346	0.334	0.302	0.263		
	taud		2.230	2.306	2.367	2.320	2.398	2.451	2.411	2.420	2.493	2.435	2.301	2.162		
	Ebn at Noon		174	225	257	264	274	279	269	261	252	218	164	137		
	Edh at Noon		15	22	28	34	34	32	33	31	24	20	14	11		
(44) All-Sky Solar Radiation	RadAvg		107	286	770	1241	1639	1852	1655	1303	882	396	140	73		
	RadStd		19	49	75	155	141	100	205	115	104	63	27	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page